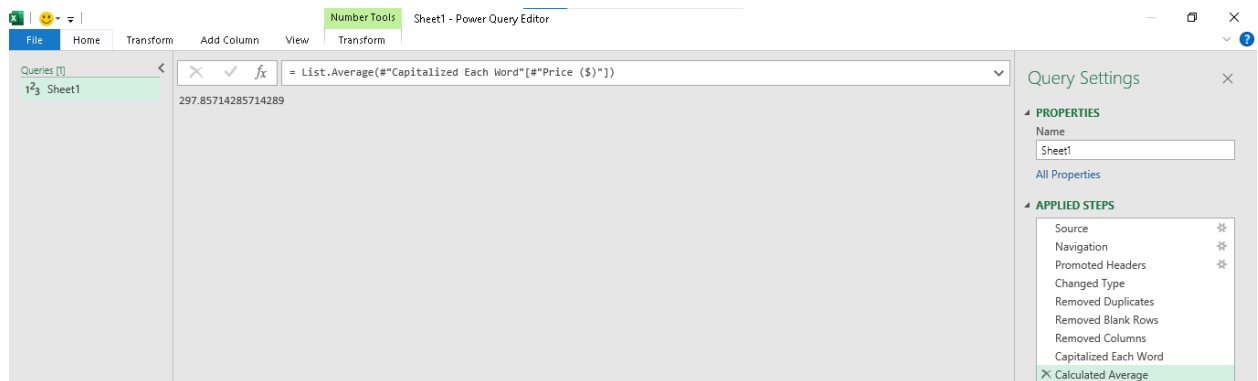


1. Handling Missing Values:

(I) Check for missing values in the 'Price' column. How would you handle products with missing price information?

Steps Involved - Firstly selected the price column and navigated to “Transform Tab” and then to Statics tab to find the average price. For which I got the value of 298.



(ii) If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

Steps Taken - After finding the average value, it is replaced with null.

The screenshot shows the Power Query Editor interface. The formula bar at the top displays the formula: `= Table.ReplaceValue(#"Capitalized Each Word",null,298,Replacer.ReplaceValue,{"Price ($)"})`. Below the formula bar, a table of data is shown. The 'Price (\$)' column is highlighted in green. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list. The steps are: Source, Navigation, Promoted Headers, Changed Type, Removed Duplicates, Removed Blank Rows, Removed Columns, Capitalized Each Word, and Replaced Value. The 'Replaced Value' step is highlighted.

	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	null
6	07-JUN-UK	Headphones	Sony	298	45	Electronics
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	null
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics

2. Correcting Inconsistent Data:

(I) Identify any inconsistent text formats present in the "Product Name" column. • Identify any typos present in the "Category" column.

Steps Taken - Product Name was in inconsistent format. Hence, selected the Product name, then navigated to “Transform” and then selected Capitalize Each Word.

	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Column7	Column8
1	28-JAN-US	laptop	Dell	1000	30	Electronics		
2	15-FEB-US	Sneakers	Nike	80	15	Fashion		
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen		
4	11-APR-US	smartphone	Samsung	900	25	Electronics		
5	22-MAY-US	Backpack	North Face	70	20	null		
6	07-JUN-UK	Headphones	Sony	null	45	Electroni		
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion		
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen		
9	05-SEP-UK	Tablet	Apple	500	50	Electronics		
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor		
11	17-11-IN-IN	Laptop	HP	450	75	Electronics		

Typos in Category column is replaced with right words “Electroni” is replaced with right word of “Electronics”

3. Removing Duplicates:

Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

Steps Taken – Identified and duplicates have been removed in the initial stage of the process.

= Table.RenameColumns("#Removed Columns2",{{"Merged", "Category"}})						
	1 ^A _C Product ID	1 ^A _C Product Name	1 ^A _C Brand Name	1 ² Price (\$)	1 ² Quantity	1 ^A _C Category
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	FashionFashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	KitchenElectronics
4	11-APR-US	Smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	Outdoor
6	07-JUN-UK	Headphones	Sony	298	45	Electronics
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	Fashion
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	Electronics
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	298	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Outdoor
24	29-SEP-CA	Smartphone	Google	800	45	Electronics
25	03-JUN-CA	Sunglasses	Ray-Ban	298	25	Fashion

Query Settings

PROPERTIES
Name
Assignment 2
All Properties

APPLIED STEPS

- Source *
- Navigation *
- Promoted Headers *
- Changed Type
- Removed Duplicates
- Removed Blank Rows
- Removed Columns
- Capitalized Each Word
- Replaced Value *
- Replaced Value1 *
- Replaced Value2 *
- Duplicated Column
- Added Conditional Column *
- Removed Blank Rows1
- Removed Columns1
- Inserted Merged Column *
- Removed Columns2
- Renamed Columns**

4. Splitting and Merging Data:

(I) Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

Steps Taken - To get the desired results, firstly navigated to “ Split Column” then “By Delimiter”. Then all the result received as Date, Month and Country Code Separately.

	1 ² 3 Product ID.1	1 ^A _C Product ID.2	1 ^A _C Product ID.3
1	28	JAN	US
2	15	FEB	US
3	3	MAR	US
4	11	APR	US
5	22	MAY	US
6	7	JUN	UK
7	19	JUL	UK
8	23	AUG	UK
9	5	SEP	UK

Then merged both Date & Month column together to get the desired result and then named it as Manufacturing Date and County Code respectively.

	Manufacturing Date	Country Code
1	1/28/2026	US
2	2/15/2026	US
3	3/3/2026	US
4	4/11/2026	US
5	5/22/2026	US
6	6/7/2026	UK
7	7/19/2026	UK
8	8/23/2026	UK
9	9/5/2026	UK
10	10/14/2026	UK
11	6/17/2026	IN
12	11/25/2026	AU

(ii) Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

Steps Taken – Again used the same merging option by selecting “Brand Name” & “Product Name” then to remove the space in between the names, navigated to Replace values and then enabled the space to get the desired result.

Product Brand
LaptopDell
SneakersNike
Coffee MakerKeurig
SmartphoneSamsung
BackpackNorth Face
HeadphonesSony
T-ShirtAdidas
BlenderNinja

Product Brand
Laptop Dell
Sneakers Nike
Coffee Maker Keurig
Smartphone Samsung
Backpack North Face

5. Number Formatting:

(I) Format the data type of the "Price" column to a currency format.

Steps Taken - Selected Price column and then selected “Change Type” and then opted for “ Currency “

 Price (\$)
1,000.00
80.00
130.00
900.00
70.00
298.00
30.00
90.00
500.00
130.00

(II) Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY "

Steps Taken - After merging the date and month, I have selected the manufacturing date column and then navigated to “ Change Type” and then opted for “Using Locale” then selected “ English – United Kingdom” to get the desired result.

 Manufacturing Date
1/28/2026
2/15/2026
3/3/2026
4/11/2026
5/22/2026
6/7/2026
7/19/2026

	 A ^B _C Manufacturing Date 
1	28/01/2026
2	15/02/2026
3	03/03/2026
4	11/04/2026
5	22/05/2026
6	07/06/2026
7	19/07/2026
8	23/08/2026
9	05/09/2026

6. Conditional Formatting:

(I) Apply data bar or color scales conditional formatting in the "Price" column. ● Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

Steps Taken - After closing and loaded the data to an excel sheet, I have created an option called “ Custom formatting to “ Price Column” opted only data and then clicked “Conditional Formatting” and then opted for Data Bars.

Secondly, for Electronics under Category column, followed the similar steps as above and opted for “Color Scales “ to highlight only Electronics.

Assignment 2 - Excel						
File	Home	Insert	Draw	Page Layout	Formulas	Data
Clipboard	Font	Alignment	General			
A1	Manufacturing Date					
	A	B	C	D	E	F
1	Manufacturing Date	Country Code	Product Brand	Price (\$)	Quantity	Category
2	28/01/2026	US	Laptop Dell	1000	30	Electronics
3	15/02/2026	US	Sneakers Nike	80	15	Fashion
4	03/03/2026	US	Coffee Maker Keurig	130	40	Kitchen
5	11/04/2026	US	Smartphone Samsung	900	25	Electronics
6	22/05/2026	US	Backpack North Face	70	20	Outdoor
7	07/06/2026	UK	Headphones Sony	298	45	Electronics
8	19/07/2026	UK	T-Shirt Adidas	30	5	Fashion
9	23/08/2026	UK	Blender Ninja	90	35	Kitchen
10	05/09/2026	UK	Tablet Apple	500	50	Electronics
11	14/10/2026	UK	Hiking Boots Timberland	130	10	Outdoor
12	17/06/2026	IN	Laptop HP	950	25	Electronics
13	25/11/2026	AU	Sneakers Adidas	90	40	Fashion
14	08/12/2026	DE	Coffee Maker Nespresso	120	35	Kitchen
15	18/02/2026	CA	Smartwatch Fitbit	150	15	Electronics
16	16/04/2026	ES	Headphones Bose	250	20	Electronics
17	21/08/2026	CA	Laptop Bag Samsonite	50	35	Accessories
18	20/08/2026	CN	Smartwatch Huawei	160	15	Electronics
19	27/01/2026	IT	Laptop Asus	980	10	Electronics
20	01/03/2026	UK	Sunglasses Oakley	150	15	Fashion
21	14/08/2026	US	Camping Tent Coleman	298	10	Outdoor
22	14/05/2026	RU	Camera Nikon	700	50	Electronics
23	09/01/2026	CA	Microwave Panasonic	80	20	Kitchen